# Milestone 3

**Opened:** Wednesday, 7 October 2026, 12:00 AM
**Due:** Monday, 19 October 2026, 12:00 AM

## 1. Objective

Milestone 2 proved that your system *runs*. Milestone 3 proves that *people can use it*: every P0 story works from start to finish through your interface, with real data from the database, and every error gives the user a clear message.

> **What "done" means in M3.** A P0 story is done when a person who has never seen your app can complete it **through the interface**, without help. A story that only works when you type the URL by hand, or only when nothing goes wrong, is not done yet.

## 2. What to write

One new file: `docs/ui.md` in your team repository. Markdown only. Wireframes and screenshots go in `docs/images/ui/`.

`docs/SETUP.md` from M2 must still work — update it whenever you add a library, a setting or a test account.

Four sections, in this order. Numbers in bold are minimums.

| Section | Minimum | What it must contain |
|---|---|---|
| **1. P0 stories and status** | **every P0 story** | A table: **Story · Screen(s) · Route · Status · Issue**. Status is one of *works*, *partly works*, *not started*. Add the updated screen flow diagram from M1. |
| **2. Wireframes** | **every P0 screen 1 screen in 4 states** | One wireframe per P0 screen. Photos of paper are fine. Each one says **"The user comes here to ..."** and lists the API endpoints it calls. At least one screen is drawn in all **four states**: empty, loading, error, data. |
| **3. Error messages** | **4 messages** | A table: **API error code · Business rule · The exact message the user sees**. Use the error codes from your M2 API table. A good message says **what happened, why, and what to do next**. "Error 422", "Something went wrong" or a stack trace on screen are not acceptable. |
| **4. What changed since M2** | **1 change** | Something in `docs/design.md` — the API, the data model, the architecture — that building the interface made you change, and why. Update `design.md` in the same pull request. |

> **The rule that decides most of your mark**
>
> At the Sprint 3 Review session, the **instructor clones your repository onto a machine that has never seen your project**, follows `docs/SETUP.md`, and uses your app.

| Check | Pass condition |
|---|---|
| Clone, follow `docs/SETUP.md`, run | **No errors**, and the database is created and seeded |
| Complete one P0 story | done **through the interface**, without typing a URL by hand |
| Break one business rule on purpose (for StudyRoom: book 4 hours) | a **clear message** on screen, not "Error 422" |
| Open `docs/ui.md` | all four sections are there, in order |

> **Help the instructor find things.** On the first screen of your README, list your P0 stories and where each one starts. If your app needs a login, put **test accounts** in `SETUP.md` — never real passwords.

# 3. Worked examples

> Same example topic as M1 and M2 — **StudyRoom, booking a study desk in the library**.

## 3.1 P0 stories and status

| Story | Screen(s) | Route | Status | Issue |
|---|---|---|---|---|
| US01 Find a free desk | Availability | `/availability` | works | #41 |
| US02 Book a desk | Book | `/book` | works | #42 |
| US03 See and cancel my booking | My bookings | `/my-bookings` | **partly works** — cancel has no confirmation yet | #48 |
| US04 Check in at the desk | My bookings | `/my-bookings` | works | #45 |

## 3.2 Wireframe in four states

```
/my-bookings - "The user comes here to see tonight's booking and check in."
Calls: GET /api/my-bookings, POST /api/bookings/<id>/checkin

+- Empty ----------------------+   +- Loading --------------------+
| You have no bookings yet.    |   | (spinner) Loading...         |
| [ Book a desk ]              |   | [==========          ]       |
+------------------------------+   +------------------------------+
+- Error ----------------------+   +- Data -----------------------+
| We can't reach the server.   |   | A12 - Floor 2 - 18:00-20:00  |
| [ Try again ]                |   | [ Check in ]  [ Cancel ]     |
+------------------------------+   +------------------------------+
```

A photo of the same four boxes drawn on paper is just as good.

## 3.3 Error messages

| API code | Rule | The user sees |
|---|---|---|
| `409` | BR1 | "You already have an active booking. Cancel it before making a new one." |
| `422` | BR2 | "A booking can last at most 3 hours. Choose an end time before 21:00." |
| `403` | BR5 | "Booking is paused until 12/10 because of three no-shows this month." |
| `422` | BR4 | "Check-in closed at 18:15, so this booking became a no-show." |

## 3.4 What changed since M2

> **New endpoint `GET /api/my-bookings`.** The M2 API table had no way to list a student's own bookings — we only noticed when we drew the My bookings wireframe. Added to section 3 of `design.md` in PR #50, with its error code `401` when nobody is logged in.

# 4. Required state of your repository

| Item | Minimum |
|---|---|
| Merged pull requests in Sprint 3 | **4**, each reviewed by someone else |
| Issues closed in Sprint 3 | **5**, including the P0 stories marked *works* |
| Per member: merged PR · review given · issue closed | **≥1 each** |
| `docs/SETUP.md` — still works on a new machine | "Tested by" line updated in Sprint 3 |
| README — first screen | P0 stories and where each starts; test accounts if there is a login |
| `[Chore] Refine backlog for Sprint 3` → **PO**, closed | **1** |

| | |
|---|---|
| `[Chore] Sprint 3 wrap-up` → **SM**, closed | **1** |
| `docs/sprint-log.md` — Sprint 3 section | committed · completed · velocity |
| Scrum Master — a **different person** from Sprints 1 and 2 | named in README |
| *Optional* — `docs/retro.md`, Sprint 3 retrospective | *one action, one owner* |

> **Minimum bar**
>
> 1. PDF submitted by the deadline with the full cover block
> 2. `docs/ui.md` has all four sections, in order
> 3. `docs/sprint-log.md` records committed, completed and velocity for Sprint 3
> 4. The project board opens from the link you submitted
> 5. `docs/SETUP.md` is still on `main`

## 5. Submission workflow

1. One member creates a branch, e.g. `docs/m3-ui`
2. Commit `docs/ui.md` and the images in `docs/images/ui/` on that branch
3. Open a Pull Request into `main`
4. **Another member reviews and approves**, leaving at least one substantive comment
5. Merge the Pull Request into `main`
6. Export the merged `ui.md` to PDF and submit on the LMS

Use the **Review changes** button, not the comment box at the bottom of the page — a plain comment is not recorded as a review.

## 6. What to submit here

One file. File name: `Team<NN>_M3.pdf` — for example `Team03_M3.pdf`

```
Team:           Team 03 — <your product name>
Topic:          <your assigned topic>
Members:        Nguyen Van A (ID), Tran Thi B (ID), ...
Product Owner:  @github-username
Scrum Master:   @github-username   (Sprint 3 — must differ from Sprints 1 and 2)

Repository:     https://github.com/<org>/<repo>;
Project board:  https://github.com/orgs/<org>/projects/<n>;
Setup guide:    https://github.com/<org>/<repo>/blob/main/docs/SETUP.md

Submitted by:   Nguyen Van A
```

**Page 2 — two screenshots of your project board**, one right after Sprint Planning and one on the day you submit.

**Page 3 — two screenshots of the running app**: one P0 screen with real data, and one error message shown to the user. The browser address bar must be visible.

**If the PDF and the repository disagree, the repository is authoritative.**

Add submission

## Submission status

| | |
|---|---|
| **Submission status** | No submissions have been made yet |
| **Grading status** | Not graded |
| **Time remaining** | 1 day 1 hour remaining |

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